# Steve (Kai) Fan

Limited Term Assistant Professor · University of Georgia

## Contact

Department of Mathematics
University of Georgia
Boyd Research and Education Center
Athens, GA 30602

Steve.Fan@uga.edu
stvfan.github.io
Office: 321C Boyd

## Research Interests

**Analytic number theory:** the distribution of primes, zeta and $L$-functions, arithmetic functions, sieve methods, with applications to arithmetic statistics

**Additive combinatorics:** additive structure of arithmetic sets, including Sárközy-type problems and completeness of sets and sequences

## Education

Nov 2023 — **Ph.D. in Mathematics, Dartmouth College**
Advisor: Carl Pomerance.
Dissertation: *Rough Numbers and Variations on the Erdős–Kac Theorem*.

Feb 2020 — **M.A. in Mathematics, Dartmouth College**

Jun 2015 — **B.S. in Mathematics, Southeast University**

## Academic Appointments

Jan 2025–present — **Limited Term Assistant Professor**
Department of Mathematics, University of Georgia.
Postdoctoral mentor: Paul Pollack.

Jan 2024–Nov 2024 — **Postdoctoral Fellow**
Max Planck Institute for Mathematics, Bonn, Germany.

## Publications and preprints

[1] **The optimal van der Corput property for squares**
Steve Fan and Andrew Lott
Preprint (2026). arXiv:2610.07765v2.

## Publications and preprints (continued)

[2] **Rank stability of elliptic curves over $\mathbb{F}_q(t)$ in residue classes**
Steve Fan and Sun Woo Park
Preprint (2026). arXiv:2609.33792.

[3] **Strongly complete sets and a conjecture of Erdős**
Steve Fan
Preprint (2026). arXiv:2607.14071.

[4] **The van der Corput property for sums of two squares**
Steve Fan and Andrew Lott
Preprint (2026). arXiv:2606.29185.

[5] **On the asymptotic density of the ordered pairs $(a, b)$ of positive integers such that $\gcd(ab, a+b) = \gcd(a, b)$**
Steve Fan and László Tóth
Preprint (2026). arXiv:2606.20057v3.

[6] **Extensions of the Furstenberg–Sárközy theorem via the arithmetic level-$d$ inequality**
Carlo Francisco E. Adajar, Rishika Agrawal, Mukul Rai Choudhuri, Chian Yeong Chuah, Steve Fan, Swaroop Hegde, Andrew Lott, Krishnamohan Nandakumar, and Nagendar Reddy Ponagandla
Preprint (2026). arXiv:2605.16216.

[7] **On a super telescoping sum representing binomial coefficients**
Steve Fan
*Rocky Mountain J. Math.* **56**(1) (2026), 137–143.

[8] **A family of analogues to the Robin criterion**
Steve Fan, Mits Kobayashi, and Grant Molnar
*Ramanujan J.* **70** (2026), Article 40, 61 pp.

[9] **Sárközy's theorem in $\mathbb{F}_q[t]$ via the van der Corput property**
Steve Fan and Andrew Lott
*Finite Fields Appl.* **116** (2026), 40 pp.

[10] **The maximal order of the shifted-prime divisor function**
Steve Fan and Paul Pollack
*Integers* **26A** (2026), A9, 14 pp.

[11] **The Hardy–Ramanujan inequality for sifted sets and its applications**
Steve Fan
Preprint (2025). arXiv:2508.06005.

[12] **Counting primes with a given primitive root, uniformly**
Steve Fan and Paul Pollack
*Mathematika* **71**(4) (2025), e70055, 30 pp.

[13] **Extremal elasticity of quadratic orders**
Steve Fan and Paul Pollack
In *The ideal theory and arithmetic of rings, monoids, and semigroups*, Scott T. Chapman (ed.), Contemp. Math. 836, Amer. Math. Soc., Providence, RI, 2026, pp. 35–53.

## Publications and preprints (continued)

[14] **The typical elasticity of a quadratic order**
Steve Fan and Paul Pollack
arXiv:2411.09063 (2024). Revised manuscript under review at *Math. Z.*

[15] **The shifted prime-divisor function over shifted primes**
Steve Fan
*Ramanujan J.* **66** (2025), Article 44, 46 pp.

[16] **Shifted-prime divisors**
Steve Fan and Carl Pomerance
arXiv:2401.10427 (2024). To appear in a Springer volume dedicated to Helmut Maier on his 70th birthday, edited by J. Friedlander, C. Pomerance, and M. Rassias.

[17] **Weighted Erdős–Kac theorems via computing moments**
Steve Fan
*Acta Arith.* **217**(2) (2025), 99–158.

[18] **Numerically explicit estimates for the distribution of rough numbers**
Steve Fan
*J. Number Theory* **260** (2024), 120–150.

[19] **An inequality related to the sieve of Eratosthenes**
Steve Fan and Carl Pomerance
*J. Number Theory* **254** (2024), 169–183.

[20] **An inequality for the distribution of numbers free of small prime factors**
Steve Fan
*Integers* **22** (2022), A26, 12 pp.

[21] **The second largest Balaban index (sum-Balaban index) of unicyclic graphs**
Wei Fang, Yubin Gao, Kai Fan, and Zhongshan Li
*J. Math. Res. Appl.* **37**(4) (2017), 391–403.

[22] **A finite difference scheme for semilinear space-fractional diffusion equations with time delay**
Zhaopeng Hao, Kai Fan, Wanrong Cao, and Zhizhong Sun
*Appl. Math. Comput.* **275** (2016), 238–254.

## Recent Talks

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|---|---|
| Sept 2026 | **The van der Corput property for sums of two squares**<br>Number Theory Seminar, Duke University. |
| May 2025 | **Counting shifted-prime divisors**<br>Plenary talk, Integers Conference 2025. |
| Mar 2025 | **The typical elasticity of a quadratic order**<br>Number Theory/Algebraic Geometry Seminar, University of Georgia. |
| Nov 2024 | **The typical elasticity of a quadratic order**<br>The Fouvry Seminar, University of Paris-Saclay, Orsay, France. |

## Recent Talks (continued)

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|---|---|
| Oct 2024 | **The typical elasticity of a quadratic order**<br>Number Theory Lunch Seminar, Max Planck Institute for Mathematics. |
| May 2024 | **Counting shifted-prime divisors**<br>Intercity Number Theory Seminar, Utrecht University. |

## Teaching

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| **University of Georgia** | Instructor of record |
| Fall 2026 | MATH 4400/6400: Number Theory |
| Spring 2026 | MATH 2700: Differential Equations |
| Fall 2025 | MATH 2250: Calculus I<br>Two sections |
| Spring 2025 | MATH 1113: Precalculus<br>Two sections |
| **University of Georgia** | Directed Reading Program Mentor |
| Spring 2026 | Roth's theorem on three-term arithmetic progressions<br>Mentee: Rishab Kalyan Subramaniyan |
| **Dartmouth College** | Instructor of record |
| Fall 2022 | MATH 11: Accelerated Multivariable Calculus |
| Fall 2021 | MATH 1: Introduction to Calculus |
| Summer 2020 | Math Camp: Exploring Mathematics |
| **Dartmouth College** | Teaching assistant |
| Spring 2020 | MATH 13: Multivariable Calculus |
| Fall 2019 | MATH 23: Differential Equations |
| Winter 2019 | MATH 3: Calculus |
| Fall 2018 | MATH 3: Calculus |